SMART INDIA HACKATHON 2026

Problem Statement ID	**SIH26102**
Problem Statement Title	**Development of an AI-powered system to detect anomalies, fraud, and inefficiencies in MPLAD Scheme implementation regd.**
Organisation	**Ministry of Statistics & Programme Implementation (MoSPI)**
Theme	**Smart Automation**
PS Category	**Software**
Team ID	**121911**
Team Name	**NeuralNexus**

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DRISHTI

The AI-eye on every MPLADS rupee

Detect anomalies. Explain risks. Enable smarter oversight of public funds.

PROPOSED SOLUTION

DRISHTI is an AI-powered risk-intelligence layer on top of **eSAKSHI**. **12 AI agents** check every MPLADS work at every stage — before sanction, during construction, on money & time, and after completion — reading not just the records but the **certificates, bills and geo-tagged photos** already on eSAKSHI, and turn what they find into a **daily inspection plan** for District Authorities.

DETAILED EXPLANATION

- **12 specialist agents:** each checks one kind of problem CAG audits have found — cost, duplicates, delays, fake evidence.
- **Reads the evidence:** completion certificates, bills, Hindi utilisation certificates and geo-tagged photos.
- **Explainable Risk Score (0–100):** fuses all agents + top-3 reasons in plain Hindi/English.
- **Inspection Planner + Field App:** picks works for mandatory inspections, routed; verdicts retrain the agents.

HOW IT ADDRESSES THE PROBLEM

- **Manual monitoring can't scale →** every work screened automatically, every day.
- **Money moves before checks →** pre-sanction check inside the 45-day window.
- **Evidence uploaded, never checked →** every photo & bill read, compared and placed on the map.
- **Many alerts, few officers →** risk-ranked queue + optimised visit route.

INNOVATION & UNIQUENESS

- **Reads eSAKSHI's own evidence:** spots reused photos, wrong locations, missing plaques and bills that don't add up.
- **Proven, then measured:** methods India already runs at scale (PMAY-G, Amrit Sarovar); each agent must beat a baseline first.
- **Clause-cited, human-in-the-loop:** every flag cites the guideline; flags for review, never accuses.
- **Learns from the field:** officer verdicts become labels, solving the “no fraud labels” problem.

12 AI AGENTS ACROSS THE LIFE OF A WORK — EVERY PS REQUIREMENT COVERED

BEFORE SANCTION

Eligibility — works the rules don't allow
Duplicate & Split — funded twice, or split up
Cost — cost overruns & padded estimates

DURING CONSTRUCTION

Site Progress — claim vs photo, plaque check
Photo Forensics — reused or edited photos
Geo-Location — photo taken elsewhere
Document — bills that don't add up
Satellite — ghost ponds & long roads

MONEY & TIME

Delay Prediction — likely to miss 12 months
Fund Flow — idle funds, year-end rush
Contractor Network — a few firms win all

AFTER COMPLETION

Citizen Verification — asset exists & is used

Then, for every work:
Risk Fusion · Inspection Planner · Ask DRISHTI

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TECHNICAL APPROACH

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SYSTEM ARCHITECTURE & WORKFLOW

01 — Data Ingestion
- 2.1 lakh eSAKSHI works
- Certificates, bills & photos
- Field app & citizen uploads

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02 — Data Validation
- Date & amount checks
- Read documents & photo stamps
- Mask names & phone numbers

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03 — 12 AI Agents
- Before sanction (3)
- During construction (5)
- Money & time (3), after (1)

→

04 — Risk Fusion
- Weighted fusion → 0–100
- Top-3 reasons + evidence
- Low / Med / High / Critical

→

05 — Alerts & Dashboards
- 4 role-based views
- Inspection plan + field app
- SMS / email / PDF brief

12 AI AGENTS — HOW EACH ONE WORKS

Eligibility
How: guideline rules + AI that reads the clause
Catches: works the guidelines don't allow

Duplicate & Split
How: DIGIPIN map grid + Hindi & English text match
Catches: same work funded twice, or split up

Cost
How: cost range of similar works in the district
Catches: estimates far above similar works

Site Progress
How: vision AI checks each claim against photos
Catches: “roof done” with no roof; no plaque

Photo Forensics
How: image fingerprints across all works
Catches: same photo or certificate reused

Geo-Location
How: photo GPS & stamp vs district map
Catches: photos taken elsewhere, or too early

Document
How: AI reads certificates, bills & UCs
Catches: amounts & dates that don't add up

Satellite
How: Sentinel-2 before/after, big works only
Catches: ponds & long roads never built

Delay Prediction
How: model trained on real completion history
Catches: works likely to miss 12 months

Fund Flow
How: MP & district money funnel over time
Catches: idle funds, year-end rush

Contractor Network
How: contractor–agency–district graph
Catches: a few firms winning most works

Citizen Verification
How: QR code at site → photo + GPS
Catches: assets missing or not in use

TECHNOLOGY STACK

Frontend	Next.js (React), Tailwind CSS, MapLibre maps
Backend / APIs	Python FastAPI, role-based access (JWT)
AI / ML	LightGBM, scikit-learn, sentence-transformers
Vision & Documents	Vision AI (Gemini; open-weight option), OpenCV
Maps & Satellite	LGD boundaries, DIGIPIN, Sentinel-2
Data & Security	PostgreSQL + pgvector, hash-chained audit log

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FEASIBILITY AND VIABILITY

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FEASIBILITY ANALYSIS

Technical Feasibility

Proven Methods
Every agent reuses a method India already runs at scale: PMAY-G geo-tagged photos, Amrit Sarovar ponds.

Data Feasibility

Real eSAKSHI Data
2,09,993 works already loaded; 2 in 3 completed works carry certificates or photos to check.

Operational Feasibility

Workflow Fit
Read-only layer on eSAKSHI; fits the MP → District → Agency flow, no new data entry.

Economic Feasibility

Low Cost
Open-source tools; pay-per-use AI at well under ₹1 a document; no licence fees.

CHALLENGES & RISKS

Challenges & Risks	Strategies to Overcome
No Confirmed Fraud Labels Few proven cases to learn from.	**Real Outcomes as Labels** Completion history & planted tests; officer verdicts retrain.
False Alerts Terrain, rates or bulk orders can look like overpricing.	**Independent Checks Must Agree** High risk needs separate agents to agree.
Handwritten & Hinglish Records Hindi certificates and messy text are hard to read.	**Confidence Gate** Unclear readings become “can't tell”, never an alert.
Sensitive Data Named officials, contractors and personal details.	**Fair & Secure by Design** “Needs verification”, never “fraud”; role-based access.

VIABILITY

Human Control
Human-in-the-Loop
AI flags risks; officials verify and decide.

→ **Existing Workflow**
Workflow Compatibility
Works alongside existing eSAKSHI workflows.

→ **Priority-Based Action**
Risk-Based Verification
Picks higher-risk works for mandatory inspections.

→ **Future Expansion**
Modular and Expandable
Agents added one by one; fits PMAY-G & MGNREGA.

PROOF BEFORE LAUNCH

Every agent is tested on real eSAKSHI data and must beat a simple baseline before it goes live. If it fails, it stays switched off.

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IMPACT AND BENEFITS

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IMPACT ON TARGET STAKEHOLDERS

Ministry (MoSPI)

- National risk heat-map by state & district
- Spot systemic issues & guideline gaps
- Evidence-backed reports for Parliament
- Every agent's accuracy tracked openly

State Nodal Authorities

- District & agency league tables
- Escalate stalled or high-risk works
- Track follow-up on every alert
- Spot contractor concentration

District Authorities

- Daily ranked alert queue with evidence
- Inspection plan with optimised route
- Pre-sanction check in the 45-day window
- One-click PDF evidence brief

Members of Parliament

- Live status of recommended works
- Early warning on delays & idle funds
- Citizens confirm completed assets
- Transparent record builds public trust

BENEFITS

SOCIAL

- Faster delivery of community assets
- Public trust through transparency
- Citizens get assets that truly exist
- Fairer spread of works across areas

ECONOMIC

- Flag risks before funds move
- Targeted visits save officer time
- Catch reused photos & padded bills
- Fewer cost overruns and rework

ENVIRONMENTAL

- Paperless, digital monitoring
- Fewer wasted site trips and fuel
- Durable assets built right first time
- Assets tracked through their life cycle

EXPECTED OUTCOMES

100% of works screened daily; 2,09,993 real works already done

0.4% flagged for review: a queue District Authorities can clear

2 in 3 completed works have certificates or photos DRISHTI can verify

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RESEARCH AND REFERENCES

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COMPARISON WITH EXISTING SYSTEMS

Capability	DRISHTI	eSAKSHI Portal	Manual Audit / Inspection	Generic BI Dashboards
AI anomaly detection in spending	✓	✗	✗	✗
Duplicate & split works (Hindi + English)	✓	✗	Partial	✗
Reads certificates & bills (Hindi/English)	✓	✗	Partial	✗
Spots reused or edited photos	✓	✗	✗	✗
Checks where & when photos were taken	✓	Partial	✗	✗
Guideline compliance with para cited	✓	Partial	✓	✗
Delay / overrun early warning	✓	Partial	✗	Partial
Explainable 0–100 risk score	✓	✗	✗	✗
Risk-based inspection route plan	✓	✗	✗	✗
Citizen confirmation & field learning	✓	Partial	✗	✗

KEY RESEARCH INSIGHTS

₹5 Cr
per MP per year, non-lapsable funds (MoSPI)

~₹4,000 Cr
annual outlay, run on eSAKSHI since Apr 2023 (PIB)

4,918
works without the mandatory MPLADS plaque (CAG audit, 31 districts)

₹280 Cr
in 12,006 works left incomplete (CAG audit, 71 districts)

REFERENCES

Government & Policy Sources

MPLADS Guidelines 2016 & 2023 — mplads.gov.in
eSAKSHI portal & revised fund flow — MoSPI / PIB
CAG Performance Audit of MPLADS — Report 31 of 2010-11
PMAY-G AwaasApp · Mission Amrit Sarovar — MoRD / BISAG-N

Research Papers

Perceptual image hashing — Zauner, 2010
ORB feature matching — Rublee et al., ICCV 2011
Multilingual sentence embeddings — Reimers & Gurevych, EMNLP 2020
LightGBM — Ke et al., NeurIPS 2017

Datasets & Tools

eSAKSHI public works data + uploaded certificates & photos
LGD boundaries · India Post DIGIPIN — official maps
Copernicus Sentinel-2 — free satellite imagery
Gemini vision · OpenCV · scikit-learn